

Operator-based graph linear canonical transform

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ABSTRACT

This paper proposes a novel graph linear canonical transform framework based on hyper-differential operators, referred to as OGLCT. First, the relationship between the graph fractional Fourier transform (GFrFT) and the graph Fourier transform (GFT) is analyzed, and hyper-differential operators associated with the GFT matrix are derived through solutions to the Sylvester equation. Using these operators, the GFrFT, graph scaling, and graph chirp modulation are defined, culminating in the formal definition of the OGLCT and its fundamental properties. Next, the implementation of the OGLCT is discussed, and parameter optimization techniques are explored. Finally, the application of the OGLCT for feature extraction in image processing is explored, and its effectiveness in image classification tasks is investigated. By adjusting the parameters of the OGLCT, it is demonstrated that flexible tuning significantly enhances model performance, enabling more effective and versatile applications.

1. Introduction

Graph signal processing (GSP) has emerged as a crucial field with applications in sensor networks, social networks, and biological systems [1,2]. These domains are characterized by complex, irregular data structures that pose significant challenges to traditional signal processing methods. GSP is designed to address these challenges through the utilization of graph properties for the analysis of data on irregular domains [3–5]. The rise of graph-based data structures has highlighted the necessity for efficient signal processing techniques specifically adapted to their complexity. A key advancement is represented by the adaptation of classical methods to the graph domain, such as the graph Fourier transform (GFT) [6,7], which generalizes the Fourier transform to graphs, thereby facilitating signal decomposition based on the spectral properties of the graph Laplacian [8,9].

Building on the GFT, the graph fractional Fourier transform (GFrFT) [10,11] incorporates a fractional parameter, thereby enabling interpolation between vertex and spectral domains to achieve fine-grained signal analysis. More recently, the graph linear canonical transform (GLCT) [12,13] has significantly enhanced GSP's analytical capabilities through the integration of linear canonical transformations with graph structures, enabling comprehensive signal representation. These advanced techniques-GFT, GFrFT, and GLCT-are instrumental in enhancing signal representation, filtering, and transformation, thereby addressing the complexities of graph-structured data. Their flexibility and efficacy underscore GSP's transformative role in addressing diverse challenges across disciplines [14–16].

The GLCT extends the classical linear canonical transform (LCT) [17,18] to graph-structured data, addressing the unique challenges of graph signals [12,13]. In traditional signal processing, the LCT is a versatile tool encompassing transformations like the Fourier and

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fractional Fourier transforms, widely used in time-frequency analysis, optical signal processing, and phase space analysis [19–23]. Adapting the LCT to graphs enables a robust framework for analyzing data on irregular structures. However, the interplay between graph topology and signal characteristics makes direct adaptation challenging. Existing methods treat scale transforms and chirp modulation as separate, uncoordinated steps, limiting their coherence and applicability [12,24]. To address these limitations, this paper introduces a novel GLCT framework based on hyper-differential operators, providing an integrated and efficient approach to graph signal analysis while enhancing compatibility with the complexity of graph data.

The primary motivation for defining a hyper-differential operator-based GLCT (OGLCT) is to create a comprehensive and adaptable framework for graph signal analysis [25–27]. Hyper-differential operators facilitate the systematic division of the GLCT into distinct stages, including GFrFT, graph scaling, and graph chirp modulation. By integrating hyper-differential operators, the GLCT framework can be tailored to various graph-based applications, spanning network analysis and machine learning on graphs. This approach not only extends the applicability of traditional signal processing techniques to the graph domain but also opens new avenues for the analysis and manipulation of graph signals, addressing previously insurmountable challenges.

This paper is organized as follows. Section 2 presents the fundamental theory of graph signal processing and the LCT. In Section 3, the relationship between the GFrFT and the GFT is analyzed. Hyper-differential operators associated with the GFT matrix are derived by solving the Sylvester equation. Based on these operators, the GFrFT, graph scaling, and graph chirp modulation are defined, leading to the formal definition of the OGLCT and its fundamental properties. Section 4 discusses the implementation of the OGLCT and parameter optimization techniques. Section 5 explores the application of the OGLCT in image classification tasks, focusing on feature extraction for image feature embedding in neural networks. Finally, Section 6 concludes this paper.

2. Preliminaries

2.1. Graph Fourier transform

Let $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{W}\}$ be a graph, where $\mathcal{V} = \{v_0, v_1, \dots, v_{N-1}\}$ represents a finite set of vertices, $\mathcal{E}$ denotes the set of edges, and $|\mathcal{V}| = N$. The weighted adjacency matrix $\mathbf{W}$ represents the edge weights among the graph's vertices [8]. The graph Laplacian, denoted by $\mathcal{L} = \mathbf{D} - \mathbf{W}$, is defined as a symmetric difference operator, where $\mathbf{D}$ is the diagonal degree matrix of the graph $\mathcal{G}$. The i -th diagonal element d_i of $\mathbf{D}$ equals the sum of the weights of all edges incident to vertex i . A signal $\mathbf{f} = [f(0), f(1), \dots, f(N-1)]^T \in \mathbb{R}^N$ defined on the graph $\mathcal{G}$ is interpreted as a mapping from the vertex set $\mathcal{V}$ to $\mathbb{R}$, such that $v_i \mapsto f(i)$ [10].

The adjacency matrix $\mathbf{W}$, the Laplacian $\mathcal{L}$, the row-normalized adjacency matrix $\mathbf{W}_{\text{row}} = \mathbf{D}^{-1}\mathbf{W}$, and the symmetric normalized Laplacian $\mathcal{L}_{\text{norm}} = \mathbf{I}_N - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$ are all examples of graph shift operators, collectively referred to as S . Let $S = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$, where $\mathbf{\Lambda} = \text{diag}([\tilde{\lambda}_0, \tilde{\lambda}_1, \dots, \tilde{\lambda}_{N-1}])$ denotes the eigenvalues of S , with $0 = \tilde{\lambda}_0 < \tilde{\lambda}_1 \leq \tilde{\lambda}_2 \leq \dots \leq \tilde{\lambda}_{N-1}$. The matrix $\mathbf{U} = [\mathbf{u}_0, \mathbf{u}_1, \dots, \mathbf{u}_{N-1}]$ contains the generalized eigenvectors of S as its columns. For any function $\mathbf{f} \in \mathbb{R}^N$ defined on the vertices of $\mathcal{G}$, its GFT is defined as [8]

$$\hat{\mathbf{f}} = \mathcal{F}\mathbf{f} = \mathbf{U}^{-1}\mathbf{f}. \quad (1)$$

The inverse GFT is given by $\mathbf{f} = \mathcal{F}^{-1}\hat{\mathbf{f}} = \mathbf{U}\hat{\mathbf{f}}$.

2.2. Linear canonical transform

The LCT of the signal $f(t) \in L^2(\mathbb{R})$ is defined as [21,22]

$$\mathcal{F}_{\mathbf{CM}}\{f\}(u) = \begin{cases} \int_{-\infty}^{\infty} f(u')K_{\mathbf{M}}(u', u)du' & b \neq 0 \\ \sqrt{d}e^{i(cd/2)u^2}f(du') & b = 0 \end{cases}, \quad (2)$$

$$K_{\mathbf{M}}(t, u) = \frac{1}{\sqrt{b}}e^{-\frac{i\pi}{4}}e^{i2\pi\left(\frac{a}{2b}u'^2 - \frac{1}{b}uu' + \frac{d}{2b}u^2\right)}, \quad (3)$$

where $\mathbf{M} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ denotes a parameter matrix satisfying $ad - bc = 1$, where i is the imaginary unit. The LCT can be decomposed into FrFT, scaling, and chirp modulation (CM) as [20]

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\xi & 1 \end{bmatrix} \begin{bmatrix} \sigma & 0 \\ 0 & \sigma^{-1} \end{bmatrix} \begin{bmatrix} \cos \frac{\alpha\pi}{2} & \sin \frac{\alpha\pi}{2} \\ -\sin \frac{\alpha\pi}{2} & \cos \frac{\alpha\pi}{2} \end{bmatrix}, \quad (4)$$

where $\xi = -\frac{ac+bd}{a^2+b^2}$ represents the CM parameter, $\sigma = \sqrt{a^2+b^2}$ is the scaling parameter, and $\alpha = \frac{2}{\pi} \cos^{-1}\left(\frac{a}{\sigma}\right) = \frac{2}{\pi} \sin^{-1}\left(\frac{b}{\sigma}\right)$ represents the FrFT parameter [24].

Based on the above decomposition, the LCT can be expressed in terms of a hyper-differential operator. A hyper-differential operator is characterized by embedding differential operators into the exponential term. The classical FrFT operator $\mathcal{F}_{c\alpha}$, scaling operator $\mathcal{R}_{c\sigma}$, and CM operator $\mathcal{M}_{c\xi}$ are representable in hyper-differential form as [25,27]

$$\mathcal{F}_{c\alpha} = \exp\left(-i\alpha\pi^2 \frac{\mathcal{P}_c^2 + \mathcal{Q}_c^2}{2}\right), \quad (5)$$

$$\mathcal{R}_{c\sigma} = \exp\left(-i2\ln(\sigma)\pi\frac{\mathcal{P}_c\mathcal{Q}_c + \mathcal{Q}_c\mathcal{P}_c}{2}\right), \quad (6)$$

$$\mathcal{M}_{c\xi} = \exp\left(-i2\xi\pi\frac{\mathcal{Q}_c^2}{2}\right), \quad (7)$$

where $\mathcal{Q}_c f(u) = uf(u)$ is the coordinate multiplication operator, and $\mathcal{P}_c f(u) = \frac{1}{i2\pi} \frac{df(u)}{du}$ is the differential operator. These two operators are Fourier duals, and their duality is described by the following relationships

$$\mathcal{P}_c = \mathcal{F}_c \mathcal{Q}_c \mathcal{F}_c^{-1}, \quad \mathcal{Q}_c = \mathcal{F}_c \mathcal{P}_c \mathcal{F}_c^{-1}, \quad (8)$$

where $\mathcal{F}_c$ denotes the classical Fourier operator.

The discrete LCT (DLCT) can be obtained by defining coordinate multiplication and differentiation in the discrete domain, followed by leveraging dual relationships. The hyper-differential operator-based DLCT is defined as [26]

$$\begin{aligned} \mathcal{F}_{dM} = \mathcal{M}_{d\xi} \mathcal{R}_{d\sigma} \mathcal{F}_{d\alpha} = \exp\left(-i2\xi\pi\frac{\mathcal{Q}_d^2}{2}\right) \\ \exp\left(-i2\ln(\sigma)\pi\frac{\mathcal{P}_d\mathcal{Q}_d + \mathcal{Q}_d\mathcal{P}_d}{2}\right) \exp\left(-i\alpha\pi^2\frac{\mathcal{P}_d^2 + \mathcal{Q}_d^2}{2}\right), \end{aligned} \quad (9)$$

where $\mathcal{Q}_d$ and $\mathcal{P}_d$ are the discrete analogs of the coordinate multiplication and differentiation operators, respectively.

3. Operator-based graph linear canonical transform

The foundational framework of the GLCT builds upon the Hermitian-Gaussian characteristic function, as introduced in previous studies [13]. This framework integrates three core components: the GFrFT, the graph scaling transform, and the graph CM. The GFT matrix, denoted as $\mathcal{F}$, is represented through its spectral decomposition as $\mathcal{F} = \mathbf{U}^{-1} = \mathbf{Q}\mathbf{R}\mathbf{Q}^{-1}$. Here, $\mathbf{Q}$ is the orthogonal matrix $\mathbf{Q} = [\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_{N-1}]$, and $\mathbf{R}$ is a diagonal matrix containing the eigenvalues, $\mathbf{R} = \text{diag}([r_0, r_1, \dots, r_{N-1}])$.

For a given graph signal $\mathbf{f}$, its GFrFT representation is formulated as

$$\hat{\mathbf{f}}_\alpha = \mathcal{F}_\alpha \mathbf{f} = \mathbf{Q}\mathcal{K}\mathbf{Q}^{-1}\mathbf{f}, \quad (10)$$

where $\mathcal{K}$ is a diagonal matrix defined as $\mathcal{K} = \text{diag}([\kappa_0, \kappa_1, \dots, \kappa_{N-1}])$, with $\kappa_l = r_l^\alpha$ for $l = 0, \dots, N-1$. The fractional order α is determined by $\alpha = \frac{2}{\pi} \cos^{-1}\left(\frac{a}{\sigma}\right) = \frac{2}{\pi} \sin^{-1}\left(\frac{b}{\sigma}\right)$.

To address the need for multi-scale signal analysis in complex graph structures, a scaling mechanism is introduced to enable adaptive weighting of frequency components. To incorporate scale variations, a scaling matrix is introduced, $\mathbf{S}_\sigma = \text{diag}([\sigma^{-\epsilon r_0}, \sigma^{-\epsilon r_1}, \dots, \sigma^{-\epsilon r_{N-1}}])$, where $\{r_l\}_{l=0}^{N-1}$ are the eigenvalues associated with the undilated eigenspace $\mathbf{Q}$, and ϵ represents the scaling adjustment parameter. The resulting scaled eigenspace $\mathbf{P} = [\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_{N-1}] = \mathbf{S}_\sigma \mathbf{Q}$ generates frequency-scaled basis vectors $\{\mathbf{p}_l\}$ that adaptively emphasize different spectral bands. The graph scale transformation for a signal $\mathbf{f}$ is expressed as

$$\mathbf{S}T_\sigma \mathbf{f} = \mathbf{P}\mathbf{Q}^{-1}\mathbf{f}, \quad (11)$$

where the scaling factor σ is computed as $\sigma = \sqrt{a^2 + b^2}$. This approach facilitates improved alignment with signals exhibiting both smooth variations and localized oscillations, thereby addressing the limitations imposed by fixed scaling in traditional graph Fourier transforms.

The CM defined on graphs enables effective characterization of non-stationary signals with vertex-dependent frequency variations, while also overcoming the inherent limitations of conventional GFTs in analyzing time-evolving spectral properties. A signal in the

graph spectral domain can be characterized as follows $\hat{s}_\xi(k) = \begin{cases} e^{-i\frac{\lambda_k^2}{\xi}}, & \xi \neq 0, \\ \mathbf{1}_N, & \xi = 0, \end{cases}$ where $\mathbf{1}_N$ represents an N -dimensional column vector

filled with ones. To map this signal back to the vertex domain, the inverse GFT is applied, yielding $s(n) = \mathbf{U}\hat{s}_\xi(k)$. This vertex-domain representation serves as the core CM term for graph signal processing. To facilitate CM operations on graph signals, we introduce the operator $\mathbf{Y}$, defined as $\mathbf{Y} = \text{diag}(\mathbf{U}\hat{s}_\xi(k)) = \text{diag}([\epsilon_0, \epsilon_1, \dots, \epsilon_{N-1}])$, where each ϵ_n corresponds to a component of the vertex-domain signal. The graph CM operator is then formulated as

$$\mathbf{C}M_\xi \mathbf{f} = \mathbf{Y}\mathbf{f}, \quad (12)$$

where the parameter ξ is determined by $\xi = \frac{ac+bd}{a^2+b^2}$.

The GLCT is expressed as [13]

$$\hat{\mathbf{f}}_M = \mathcal{F}_M \mathbf{f} = \mathbf{Y}(\mathbf{P}\mathbf{Q}^{-1})(\mathbf{Q}\mathcal{K}\mathbf{Q}^{-1})\mathbf{f} = \mathbf{Y}\mathbf{P}\mathcal{K}\mathbf{Q}^{-1}\mathbf{f}. \quad (13)$$

This representation integrates the CM operator $\mathbf{Y}$, the scaled eigenspace $\mathbf{P}$, and the fractional spectral components encapsulated in $\mathcal{K}$, applied sequentially to the graph signal $\mathbf{f}$. The algebraic formulation of the GLCT is given by $\hat{\mathbf{f}}_M(k) = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} f(n)\epsilon_k p_l(k)\kappa_l q_l^*(n)$, where ϵ_k , $p_l(k)$, κ_l , and $q_l^*(n)$ represent components derived from the CM, scaled eigenspace, spectral weights, and graph Fourier

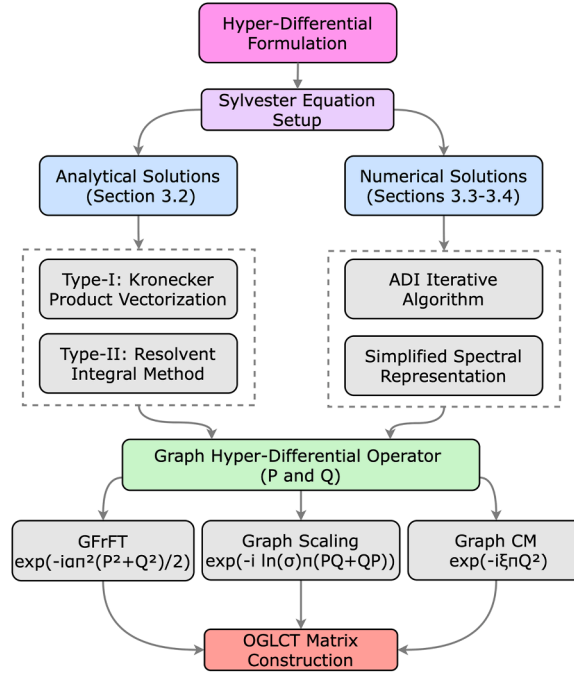


Fig. 1. Technical Workflow of the Proposed OGLCT Framework.

basis, respectively. The eigenvalues associated with the GLCT are defined as $\Lambda_{\mathbf{M}} = \text{diag}([\lambda_0, \lambda_1, \dots, \lambda_{N-1}])$, where each eigenvalue λ_l is computed as $\lambda_l = \frac{\tilde{\lambda}_l^\alpha}{\sigma}$, $l = 0, \dots, N-1$. Here, $\tilde{\lambda}_l$ denotes the original eigenvalues of the graph Laplacian, adjusted by the fractional α and the scaling factor σ .

This framework provides a comprehensive approach to extend classical signal processing concepts to graph-structured data, leveraging the spectral properties of graphs for advanced transformations and analyses. For a more comprehensive introduction, the reader is referred to our previous work [13]. While this definition provides a systematic framework for GLCT, the independent definitions of the three stages-GFrFT, graph scaling transform, and graph CM-lead to a relatively complex overall formulation. This complexity may hinder the practical utility of the transformation. Therefore, in this work, we propose a novel hyper-differential operator-based approach to define GLCT, denoted as OGLCT, aiming to streamline the formulation and enhance its utility for graph signal processing. The technical workflow of this section is illustrated in Fig. 1.

3.1. Hyper-differential forms

A primary motivation for the development of OGLCT is the establishment of a unified and versatile framework capable of accommodating various graph transformations. Within the conventional GLCT framework, the three transformation stages-GFrFT, graph scaling, and graph CM-are independently defined, each relying on distinct structural properties of the graph. Such independent definitions increase computational complexity and hinder intuitive analysis of the overall transformation. In contrast, the proposed OGLCT redefines these stages through hyper-differential operators on graphs. Once the hyper-differential operator is derived, the entire transformation framework can be constructed around it, enabling the systematic definition of all three transformation stages. This approach not only streamlines the derivation process but also offers a more coherent mathematical structure for interpreting graph signal behavior under linear canonical transformations.

Let the OGLCT operator be denoted as $F_{\mathbf{M}}$. The goal is to determine a differential operator $\mathcal{H}_\tau$ such that the OGLCT operator can be expressed in the following form [25]

$$F_{\mathbf{M}(\tau)} = \exp(-i\frac{\pi}{2}\mathcal{H}_\tau). \quad (14)$$

Here, τ corresponds to the parameter $\mathbf{M}$. The form of $\mathcal{H}_\tau$ is given by

$$\mathcal{H}_\tau = \frac{2i}{\pi} \log(F_{\mathbf{M}(\tau)}). \quad (15)$$

When the parameter $\mathbf{M} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, the GFT operator satisfies

$$\mathcal{H} = \frac{2i}{\pi} \log(F). \quad (16)$$

Building on the concept of LCT decomposition, we present the forms of hyper-differential operators for three stages: GFrFT, graph scaling, and graph CM.

The GFrFT allows for an intermediate domain between the graph vertex domain and the graph spectral domain, providing a new way to interpret and manipulate graph signals. This intermediate representation is crucial for applications that require multi-resolution analysis or where the signal's characteristics change gradually across the graph. At this stage, the objective is to express H_α as a hyper-differential operator to maintain the theoretical equivalence between the two definitions of GFrFT, where $\tau = \alpha$. Specifically, the operator H_α is defined as

$$H_\alpha = \alpha\pi(P^2 + Q^2) = \alpha\pi(P^2 + F P^2 F^{-1}), \quad (17)$$

where the graph Fourier duality between P and Q is

$$P = F Q F^{-1}, \quad Q = F P F^{-1}. \quad (18)$$

Graph scaling is crucial in the analysis and processing of graph signals, especially when adapting the signal structure to various graph topologies. Scaling transformations are essential for controlling the spread or localization of a signal within the graph domain. Consider the differential operator H_σ corresponding to the graph scaling stage, such that

$$F_{M(\sigma)} = \exp\left(-i\frac{\pi}{2}H_\sigma\right), \quad (19)$$

where $\tau = \sigma$. To determine the form of the hyper-differential operator H_σ , the operator is sought to satisfy the relationship

$$H_\sigma = 2 \ln(\sigma)(PQ + QP). \quad (20)$$

This equation reflects the coupling between the inherent structural properties of the graph, represented by the operators P and Q , and the scaling factor σ .

The graph CM stage extends the classical concept of chirp modulation to graph signal processing, enabling the modulation of graph signals in a manner that captures complex frequency components. This stage is particularly useful in applications such as graph signal filtering, where precise control over frequency characteristics is required. Given the earlier solution for P^2 in (22), the differential operator for the graph CM stage is defined as

$$H_\xi = 2\xi Q^2. \quad (21)$$

This form ensures that the modulation effect is directly related to the graph's structure, as captured by the operator Q^2 .

These operators, P and Q , represent the unknown analogs in graph signal processing that correspond to the discrete operators Q_d and P_d . Based on the relationship between GFrFT and GFT, $H_\alpha = \alpha H$ is established [24,28]. Substituting (16) into (17) yields the equation

$$F Q^2 F^{-1} + Q^2 = \frac{2i}{\pi^2} \log(F). \quad (22)$$

Therefore, to define the hyper-differential operator-based GFrFT, the objective is to solve (22) for Q^2 , given any graph Fourier matrix F . This solution ultimately enables the establishment of the desired operator. The next step is to elaborate on the procedures to obtain P and Q .

By multiplying the GFT operator F on both sides of (22), the equation becomes

$$F Q^2 + Q^2 F = \frac{2i}{\pi^2} \log(F) F. \quad (23)$$

Letting $\mathbf{A} = F$, $\mathbf{B} = F$, $\mathbf{X} = Q^2$, and $\mathbf{C} = \frac{2i}{\pi^2} \log(F) F$, (23) can be reformulated as a continuous-time Sylvester equation [29] of the form $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{B} = \mathbf{C}$, where the Sylvester equation admits a unique solution $\mathbf{X}$ if the matrices $\mathbf{A}$ and $-\mathbf{B}$ do not share any common eigenvalues [30,31]. The three methods for solving Q^2 are designed to address distinct computational needs: the explicit solutions in Section 3.2 (Kronecker vectorization and resolvent integrals) are employed to provide theoretical verification for small matrices by means of closed-form formulations, while the alternating direction implicit (ADI) method described in Section 3.3 is utilized to perform iterative large-scale computation through efficient matrix splitting. The simplified spectral approach in Section 3.4 is based on eigen-decomposition for moderate-sized systems, under the assumption of shared operator bases. This methodological spectrum is structured to balance theoretical rigor (exact solutions), practical scalability (ADI iterations), and computational efficiency (spectral simplification), thereby enabling comprehensive solutions to the Sylvester equation at varying graph scales and under diverse application requirements. These methods enable the construction of Q , and subsequently, $P = F Q F^{-1}$ is derived using graph Fourier duality.

3.2. Explicit solutions

The pursuit of explicit analytical solutions holds foundational importance for theoretical completeness and numerical validation. While often computationally intensive, closed-form solutions provide critical benchmarks for verifying the correctness of iterative methods and offer insights into the operator structure that may be obscured in numerical approximations. This section presents two analytical approaches to solve the Sylvester equation governing Q^2 : a Kronecker product-based vectorization method and a resolvent integral formulation.

3.2.1. Type-I: vectorized solution using the Kronecker product

T. Alikashioglu provides an analytical solution to (23) using the Kronecker product [28]. Building on this approach, the equation is vectorized to solve for $\mathbf{X} = \mathcal{Q}^2$. The vectorized form of the Sylvester equation is

$$(\mathcal{F} \otimes \mathbf{I} + \mathbf{I} \otimes \mathcal{F})\text{vec}(\mathbf{X}) = \text{vec}\left(\frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F}\right), \quad (24)$$

where $\otimes$ denotes the Kronecker product and $\text{vec}(\cdot)$ is the vectorization operator (i.e., $\text{vec}(\mathbf{X}) = [\mathbf{X}_{:,0}^\top, \mathbf{X}_{:,1}^\top, \dots, \mathbf{X}_{:,N-1}^\top]^\top$). Given that the Kronecker sum $\mathcal{F} \otimes \mathbf{I} + \mathbf{I} \otimes \mathcal{F}$ is invertible (assuming $\mathcal{F}$ and $-\mathcal{F}$ do not share any eigenvalues), the solution for $\text{vec}(\mathcal{Q}^2)$ can be expressed as

$$\text{vec}(\mathcal{Q}^2) = (\mathcal{F} \otimes \mathbf{I} + \mathbf{I} \otimes \mathcal{F})^{-1} \text{vec}\left(\frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F}\right). \quad (25)$$

By reshaping the vector $\text{vec}(\mathcal{Q}^2)$ back into matrix form, the solution $\mathcal{Q}^2$ for the original Sylvester equation is obtained.

3.2.2. Type-II: integral of resolvents

The second solution, referred to as Type-II, is derived using spectral theory, leveraging the residue theorem for analytic functions and the properties of the resolvent operator, as originally introduced by Krein [32], Lancaster [33]. This approach expresses the solution of the Sylvester equation in terms of resolvent integrals. Specifically, the closed-form expression for the matrix $\mathcal{Q}^2$ can be expressed as

$$\mathbf{X} = \mathcal{Q}^2 = -\frac{1}{4\pi^2} \int_{\Gamma_1} \int_{\Gamma_2} (\lambda \mathbf{I} - \mathcal{F})^{-1} \frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F}(\mu \mathbf{I} - \mathcal{F})^{-1} \frac{d\mu d\lambda}{\lambda + \mu}, \quad (26)$$

where Γ_1 and Γ_2 are contours in the complex plane enclosing the spectrum of the operator $\mathcal{F}$. The integrals utilize the resolvent operator $(\lambda \mathbf{I} - \mathcal{F})^{-1}$, which is well-defined as long as the contours are sufficiently close to the spectrum of $\mathcal{F}$.

3.3. Iterative solution through ADI

In the previous section, two analytical solutions to the problem were presented and examined. However, these solutions are often impractical or computationally intractable in real-world scenarios, particularly when dealing with large-scale or highly complex systems. In such cases, iterative numerical methods, such as the ADI method, provide an efficient framework for addressing matrix equations [34–36]. This section demonstrates the application of the ADI method to solve the Sylvester equation associated with the problem. Given the Sylvester equation

$$\mathcal{F}\mathbf{X} + \mathbf{X}\mathcal{F} = \frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F}, \quad (27)$$

the ADI method provides an efficient iterative solution by decomposing the equation into manageable steps. The procedure involves several stages, including initialization, iterative updates, and a convergence check.

Initially, the starting point is set by initializing $\mathbf{X}_0$, which is typically chosen as the zero matrix or an approximation to the solution. Additionally, iteration parameters p_k and q_k are defined for each iteration. These parameters can either be determined based on the problem's structure or set as constants. The iterative process consists of two main steps in each iteration k . In the first half-step, the intermediate variable $\mathbf{X}^{(k+\frac{1}{2})}$ is computed by solving the linear system

$$(\mathcal{F} - p_k \mathbf{I})\mathbf{X}^{(k+\frac{1}{2})} = \frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F} - \mathbf{X}^{(k)}(\mathcal{F} - p_k \mathbf{I}), \quad (28)$$

where $\mathbf{X}^{(k)}$ is the current iterate. This step provides an intermediate approximation of the solution. In the second half-step, the next iterate $\mathbf{X}^{(k+1)}$ is computed by solving the equation

$$\mathbf{X}^{(k+1)}(\mathcal{F} - q_k \mathbf{I}) = \frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F} - \left(\mathcal{F} - q_k \mathbf{I}\right)\mathbf{X}^{(k+\frac{1}{2})}. \quad (29)$$

This step updates $\mathbf{X}$ using the intermediate result $\mathbf{X}^{(k+\frac{1}{2})}$.

The process is repeated for a predetermined number of iterations K , or until convergence is achieved. Convergence is typically assessed by monitoring the residual, $\|\mathcal{F}\mathbf{X} + \mathbf{X}\mathcal{F} - \frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F}\|$, or by checking if the change in successive iterates is sufficiently small. Let the convergence precision be ϵ ; the process is considered converged when the condition

$$\|\mathcal{F}\mathbf{X} + \mathbf{X}\mathcal{F} - \frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F}\| < \epsilon, \quad (30)$$

is satisfied. After each iteration or at specified intervals, the solution can be stored for later analysis. Once convergence is achieved, the final solution $\mathbf{X}$ is output. Fig. 2 summarizes the iterative process for solving the Sylvester equation using the ADI method.

While the ADI method is an effective approach for solving Sylvester equations, several important considerations must be addressed during its application. The choice of parameters p_k and q_k is critical for balancing convergence speed and stability, as poorly chosen values may slow down or even hinder convergence. In practical applications, these parameters are typically selected based on the spectral properties of $\mathbf{A}$ and $\mathbf{B}$. In the provided implementation, they are initialized as zero vectors when unspecified; however, more effective strategies, such as spectral shift methods, may enhance convergence. Additionally, convergence is closely tied to the spectral properties of $\mathcal{F}$, which may necessitate adjusting iteration parameters or stopping criteria for specific problems.

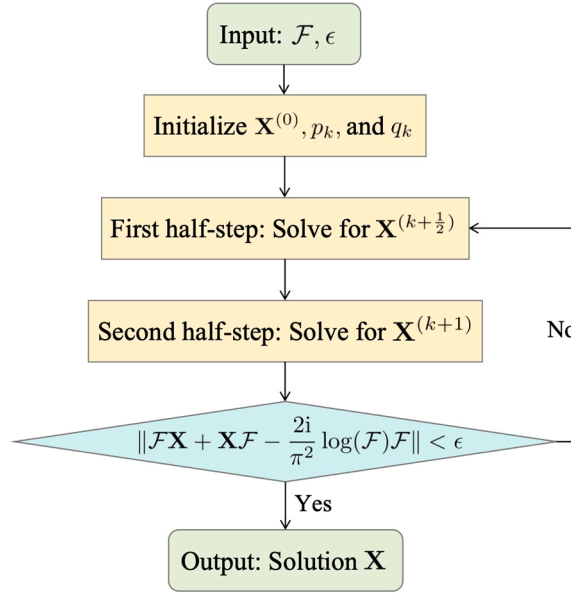


Fig. 2. The ADI iteration for (27).

3.4. Simplified matrix representation

An efficient eigenvalue-driven approach is developed in this section, which becomes particularly valuable when the graph Fourier matrix F possesses a stable spectral decomposition. By constraining Q to share the eigenbasis of F , the matrix Sylvester equation is transformed into scalar operations on eigenvalues, thereby yielding both theoretical insight and practical computational gains for moderate-sized graphs. This section focuses on the assumption that the operator Q can be represented using the eigenvector matrix of F . This assumption significantly simplifies the problem by allowing both operators to share a common spectral basis. By working in the eigenvalue space, the complexity of the equations is reduced, transforming them into scalar forms that can be solved independently for each eigenvalue. This approach not only streamlines the computation but also ensures consistency in the representation of Q within the graph Fourier transform framework.

Let $F = U^{-1} = QRQ^{-1}$, where $R = \text{diag}([r_0, r_1, \dots, r_{N-1}])$ represents the diagonal matrix of eigenvalues. Assume that Q can also be diagonalized using the same eigenvector matrix Q , such that $Q = Q\Theta Q^{-1}$, where $\Theta = \text{diag}([\theta_0, \theta_1, \dots, \theta_{N-1}])$ denotes the diagonal matrix of eigenvalues associated with Q . By substituting the decomposed forms of F and Q into (22), we obtain

$$(Q\Theta^2 Q^{-1})(QRQ^{-1}) + (QRQ^{-1})(Q\Theta^2 Q^{-1}) = \frac{2i}{\pi^2} (Q \log(R) Q^{-1})(QRQ^{-1}),$$

where $\log(R) = \text{diag}([\log(r_0), \log(r_1), \dots, \log(r_{N-1})])$.

Since $QQ^{-1} = I$, operations can be performed directly in the eigenvalue space, simplifying the expression to $\Theta^2 R + R\Theta^2 = \frac{2i}{\pi^2} \log(R)R$. For each eigenvalue component, the problem reduces to solving the following scalar equation

$$\theta_i^2 r_i + r_i \theta_i^2 = \frac{i}{\pi^2} \log(r_i) r_i, \quad (31)$$

where $i = 0, 1, \dots, N-1$. The solution for the eigenvalue θ_i is then given by

$$\theta_i = \sqrt{\frac{i}{\pi^2} \log(r_i)}. \quad (32)$$

Finally, using the reconstructed eigenvalue matrix Θ and the eigenvector matrix Q , the operator Q can be expressed in its complete form.

3.5. The OGLCT

Given the earlier solution for Q^2 derived from the Sylvester equation and its graph Fourier dual $P^2 = FQ^2F^{-1}$, these can be substituted into the hyper-differential operators of GFrFT, graph scaling, and graph CM

$$F_{M(\alpha)} = \exp\left(-i\frac{\pi}{2} H_\alpha\right) = \exp\left(-i\alpha\pi^2 \frac{P^2 + Q^2}{2}\right), \quad (33)$$

$$F_{M(\sigma)} = \exp\left(-i\frac{\pi}{2}H_\sigma\right) = \exp(-i\ln(\sigma)\pi(PQ + QP)), \quad (34)$$

$$F_{M(\xi)} = \exp\left(-i\frac{\pi}{2}H_\xi\right) = \exp(-i\xi\pi Q^2). \quad (35)$$

The hyper-differential operator-based graph scaling stage provides a robust method for controlling signal localization within the graph domain. This formulation ensures that the modulation introduced by ξ is consistent with the graph's spectral properties, enabling precise control over the frequency components of the graph signal.

Based on the above conclusions, the OGLCT matrix is defined as

$$F_M = F_{M(\xi)}F_{M(\sigma)}F_{M(\alpha)} = \exp(-i\xi\pi Q^2) \exp(-i\ln(\sigma)\pi(PQ + QP)) \exp\left(-i\alpha\pi^2 \frac{P^2 + Q^2}{2}\right), \quad (36)$$

where Q is derived from the Sylvester equation in (22), and $P = FQF^{-1}$ represents the graph Fourier dual of Q^2 . The inclusion of Q is crucial, as it preserves signal integrity across domains. This approach enhances the ability to capture complex relationships within graphs, providing a more robust and versatile tool for graph signal processing. The OGLCT for a signal $\mathbf{f}$ is expressed as (Algorithm 1)

$$\hat{\mathbf{f}}_M = F_M \mathbf{f}. \quad (37)$$

Algorithm 1 The OGLCT.

Input: GFT matrix F , parameter $M = (\xi, \sigma, \alpha)$.

Output: OGLCT matrix F_M .

- 1: Solve the Sylvester equation for Q^2 :
 $FQ^2 + Q^2F = \frac{2i}{\pi^2} \log(F)F$.
 - 2: Compute $P^2 = FQ^2F^{-1}$ as graph Fourier dual of P^2 .
 - 3: Compute the GFrFT: $F_{M(\alpha)} = \exp\left(-i\alpha\pi^2 \frac{P^2 + Q^2}{2}\right)$.
 - 4: Perform the graph scaling stage: $F_{M(\sigma)} = \exp(-i\ln(\sigma)\pi(PQ + QP))$.
 - 5: Apply the graph CM stage: $F_{M(\xi)} = \exp(-i\xi\pi Q^2)$.
 - 6: Combine stages to obtain the OGLCT: $F_M = F_{M(\xi)}F_{M(\sigma)}F_{M(\alpha)}$.
-

The proposed OGLCT fundamentally reorients this paradigm by introducing hyper-differential operators P and Q as universal graph signal processors. Intrinsic differential structures that inherently encode both graph topology and signal propagation dynamics are derived systematically via the Sylvester equation. The three transformation stages subsequently emerge naturally through the exponential mappings of these unified operators. This operator-centric approach eliminates the need for stage-specific spectral recalculations-what previously required three distinct graph-characteristic adaptations is now unified through a single derivation of P/Q , thereby ensuring mathematical consistency across all transformation phases based on their common operator foundation.

3.6. Properties

In this section, several key properties of the hyper-differential operator-based GLCT, including invertibility, additivity, and the convolution theorem, are derived.

- (a) **Invertibility:** The invertibility of the GLCT is demonstrated by taking the Hermitian conjugate of the GLCT operator, which yields

$$(F_M)^H = \exp(i\xi\pi Q^2) \exp(i\ln(\sigma)\pi(PQ + QP)) \exp\left(i\alpha\pi^2 \frac{P^2 + Q^2}{2}\right).$$

This shows that the inverse of the GLCT corresponds to another GLCT with conjugate parameters. The inverse can be expressed as $F_M^{-1} = F_{M^\Theta}$, where $M^\Theta = (-\xi, \sigma^{-1}, -\alpha)$. This indicates that the inverse GLCT is obtained by reversing the modulation, scaling, and fractional Fourier parameters, thus ensuring the transform is unitary and invertible.

- (b) **Additivity:** Two GLCT matrices are considered, defined as follows

$$F_{M_k} = \exp(-i\xi_k\pi Q^2) \exp(-i\ln(\sigma_k)\pi(PQ + QP)) \times \exp\left(-i\alpha_k\pi^2 \frac{P^2 + Q^2}{2}\right), \quad k = 1, 2. \quad (38)$$

The additivity property of the GLCT is expressed as

$$F_{M_1} F_{M_2} = F_{M_1 \oplus M_2}, \quad (39)$$

where $M_k = (\xi_k, \sigma_k, \alpha_k)$, and the composition of parameters is given by $M_1 \oplus M_2 = (\xi_1 + \xi_2, \sigma_1\sigma_2, \alpha_1 + \alpha_2)$. Here, the combination $\oplus$ represents the cumulative effect of applying two successive GLCTs and results in a new GLCT with parameters derived from the individual transforms.

(c) **Convolution theorem:** The GLCT is defined by the following operator $F_M = \exp(-i\xi\pi Q^2) \exp(-i\ln(\sigma)\pi(PQ + QP)) \exp(-i\alpha\pi^2 \frac{P^2+Q^2}{2}) = [\varsigma_0, \varsigma_1, \dots, \varsigma_{N-1}]$. The GLCT can then be expressed as $\hat{f}_M(k) = \sum_{n=0}^{N-1} f(n)\varsigma_k(n)$. The convolution in the GLCT domain is described as

$$(f * g)(n) = \sum_{k=0}^{N-1} \hat{f}(k)\hat{g}(k)\varsigma_k^*(n), \quad (40)$$

or equivalently, $\mathbf{f} * \mathbf{g} = F_M^{-1}(\hat{\mathbf{f}} \odot \hat{\mathbf{g}})$, where $\odot$ represents the Hadamard product.

4. Implementation and parameter optimization

4.1. Implementation and analysis

In this section, the implementation details of the OGLCT are described, and its performance is systematically evaluated.

4.1.1. Analysis of computational complexity

A comparative analysis of the computational complexity of the four previously introduced methods is conducted, offering insights into their respective practical applicability.

The computational complexity of the four methods for solving (23) is analyzed. The Type-I method involves solving a $N^2 \times N^2$ Kronecker system, with a worst-case complexity of $\mathcal{O}(N^6)$, rendering it impractical for large matrices. The Type-II method requires eigenvalue decomposition ($\mathcal{O}(N^3)$) and integration over N^2 eigenvalue pairs, leading to an overall complexity of $\mathcal{O}(N^5)$. Although more feasible than Type-I, it remains computationally expensive.

The ADI method iteratively solves two linear systems of size $N \times N$ in $\mathcal{O}(kN^3)$, where k represents the number of iterations. This renders it suitable for large-scale problems, particularly in cases involving sparse or well-conditioned matrices. The simplified solution is based on spectral decomposition ($\mathcal{O}(N^3)$) and lightweight eigenvalue operations, making it the most efficient for small to moderate sizes when P is well-conditioned.

To validate the theoretical complexity, experiments were performed by solving the Sylvester equation for matrices of increasing sizes N . The tested sizes ranged from $N = 10$ to $N = 200$, with each method executed 10 times per size. The average computation time was recorded to mitigate the impact of system variability. The experimental results validate the theoretical complexities of the methods, as shown in Fig. 3. For small matrices ($N < 50$), computation times are similar; however, for larger N , the Type-I and Type-II methods become impractical due to their steep growth rates ($\mathcal{O}(N^6)$ and $\mathcal{O}(N^5)$). In contrast, the ADI and simplified methods, characterized by slower growth rates ($\mathcal{O}(kN^3)$ and $\mathcal{O}(N^3)$), remain efficient and suitable for large-scale problems.

4.1.2. Analysis of numerical accuracy

The effectiveness of the OGLCT in processing graph signals is assessed, with particular emphasis on its advantages and limitations under real-world conditions. Specifically, the accuracy of the four Sylvester equation solvers was evaluated by varying the number of vertices N .

For each size $N \in \{10, 20, 30, \dots, 200\}$, a random symmetric matrix F was constructed to ensure computational stability. A ground truth solution Q_{true}^2 was randomly constructed, and the corresponding right-hand side $C = FQ_{\text{true}}^2 + Q_{\text{true}}^2 F$ was derived as the exact reference. The four numerical methods (Type-I, Type-II, ADI, and Simplified Solution) were employed to solve the Sylvester equation, producing estimated solutions Q_{est}^2 . The relative error, calculated as $\|Q_{\text{est}}^2 - Q_{\text{true}}^2\|_F / \|Q_{\text{true}}^2\|_F$, was averaged over 10 trials for each N .

As shown in Fig. 4, the Type-I method demonstrates the largest errors for larger matrix sizes ($N > 80$), primarily due to numerical instabilities exacerbated by the high computational complexity of the Kronecker product formulation. The Type-II method achieves

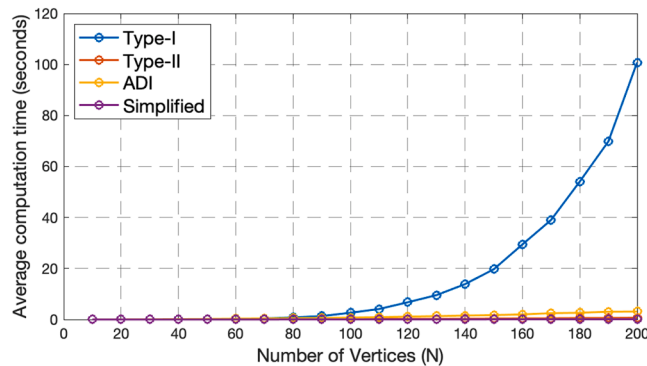


Fig. 3. Comparison of the average computation time across different methods.

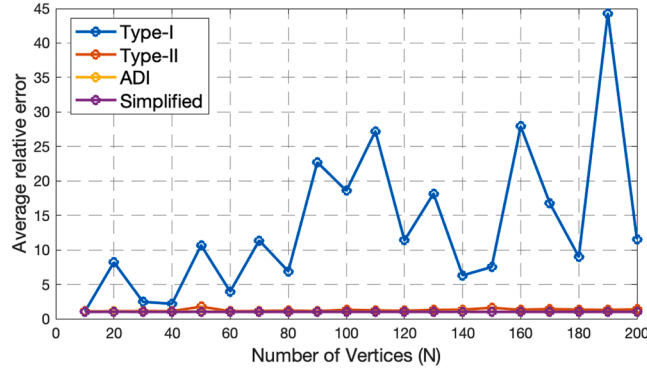


Fig. 4. Average relative error as a function of size N for the four methods.

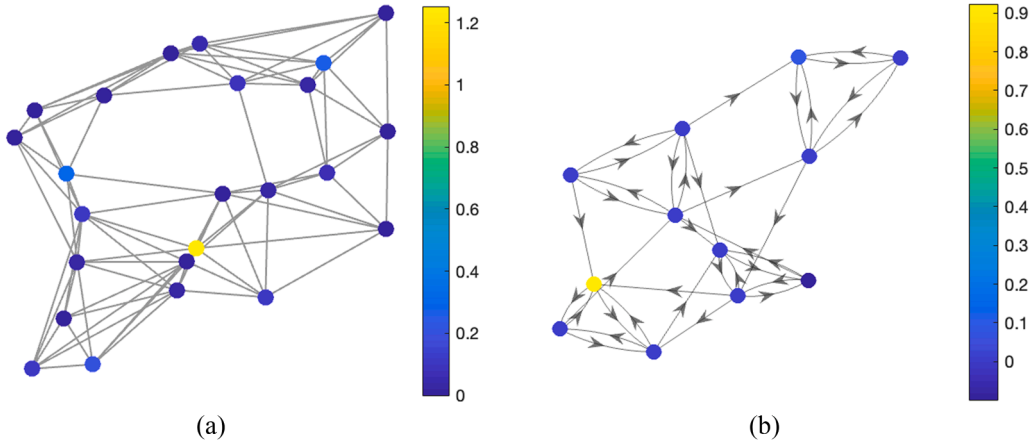


Fig. 5. Graph signals on directed and undirected graphs.. (a) Graph signal $\mathbf{f}_1 = 0.2\Re(\zeta_{11}(n)) + 3\Re(\zeta_{20}(n))$, where $\zeta_k(n)$, $k = 11, 20$ is derived from the GLCT of the undirected graph. (b) Graph signal $\mathbf{f}_2 = \Re(\zeta_{11}(n))$, where $\zeta_{11}(n)$ is derived from the GLCT of the directed graph.

better accuracy than Type-I but exhibits a moderate increase in error for $N > 160$, which is attributed to the cumulative numerical errors arising from integrating over a larger number of eigenvalue pairs. In contrast, the ADI method maintains consistently low relative errors across all tested matrix sizes, demonstrating its robustness and stability via iterative refinement. The Simplified Solution achieves the highest accuracy with negligible errors, owing to its reliance on spectral decomposition, which ensures precision provided that the matrix F is well-conditioned.

4.1.3. Illustrative implementation

A concrete implementation of the OGLCT is analyzed to illustrate the transform's operational characteristics and effectiveness. Specifically, two distinct graph signals are processed through the proposed framework.

First, an undirected graph comprising 24 vertices was analyzed. The graph signal, denoted as $\mathbf{f}_1$, was defined as $\mathbf{f}_1 = 0.2\Re(\zeta_{11}(n)) + 3\Re(\zeta_{20}(n))$, where $\Re(\cdot)$ denotes the real part. The terms $\zeta_k(n)$ for $k = 11, 20$ were derived from the OGLCT of the undirected graph, as illustrated in Fig. 5(a). Subsequently, a directed graph comprising 12 vertices was examined. The corresponding graph signal, $\mathbf{f}_2$, was expressed as $\mathbf{f}_2 = \Re(\zeta_{11}(n))$, with $\zeta_{11}(n)$ derived from the OGLCT of the directed graph, as shown in Fig. 5(b).

The OGLCT results for these graph signals, calculated using the undirected and directed graph Laplacians, are depicted in Figs. 6 and 7, respectively. When the transformation parameters were set to $\sigma = 0.5$, $\xi = 1$, and $\alpha = 0.8$, the frequency components of the graph signals were accurately resolved. This demonstrates that adjusting the OGLCT parameters can significantly enhance the spectral analysis capabilities, enabling a more accurate representation of graph signals in the frequency domain.

4.2. Parameter optimization

The previous analysis has explored the definition and implementation of the OGLCT. The OGLCT is of significant value in graph signal processing due to its versatile applications, providing a unified framework that simplifies interpretation and enhances accessibility. A key feature of the OGLCT is the incorporation of three tunable parameters, which enable more flexible representations of graph signals. Therefore, the selection of these parameters is critical to the effectiveness of the transform. To broaden its potential applications, a trainable OGLCT is proposed, with parameters optimized using the quasi-Newton method.

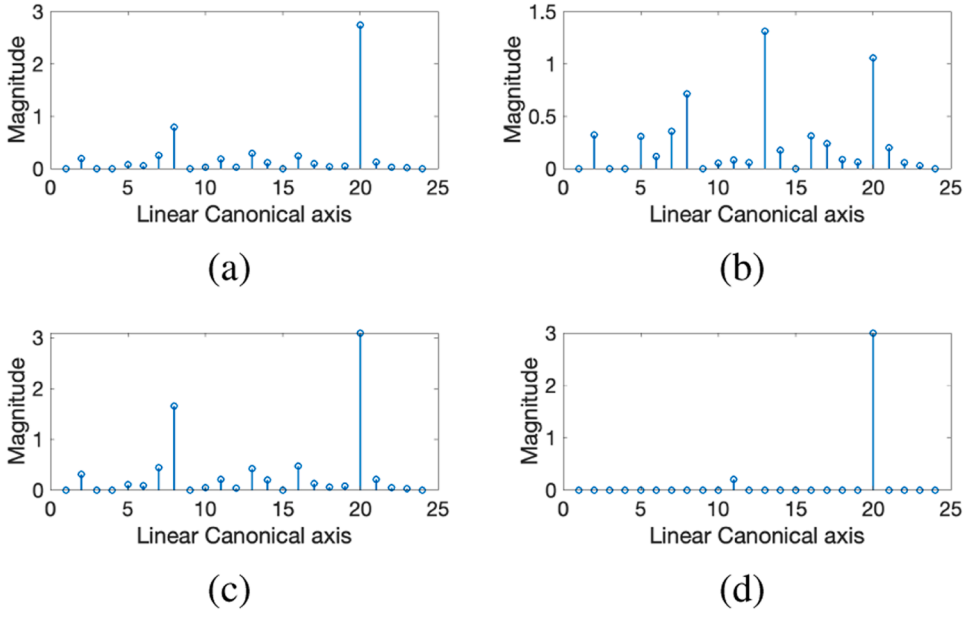


Fig. 6. Spectral graph transforms applied to graph signal $\mathbf{f}_1$. (a) GFT. (b) GFrFT with $\alpha = 0.5$. (c) GLCT with $\sigma = 1.5$, $\xi = 0.5$, $\alpha = 0.8$. (d) GLCT with $\sigma = 0.5$, $\xi = 1$, $\alpha = 0.8$.

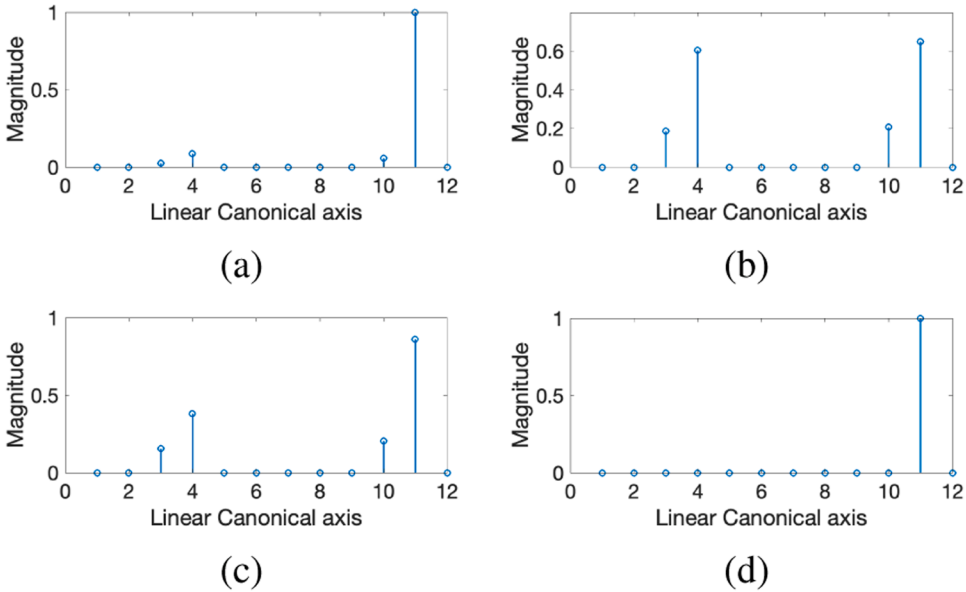


Fig. 7. Spectral graph transforms applied to graph signal $\mathbf{f}_2$. (a) GFT. (b) GFrFT with $\alpha = 0.5$. (c) GLCT with $\sigma = 1.5$, $\xi = 0.5$, $\alpha = 0.8$. (d) GLCT with $\sigma = 0.5$, $\xi = 1$, $\alpha = 0.8$.

In the context of OGLCT, the objective is to learn the transform parameters $\mathbf{M} = (\xi, \sigma, \alpha)$. Given a known graph shift operator $S \in \mathbb{R}^{N \times N}$, the graph signal is denoted as $\mathbf{f} \in \mathbb{R}^{N \times M}$. Starting with initial ideal parameters $\mathbf{M}_{\text{ideal}} = (\xi_{\text{ideal}}, \sigma_{\text{ideal}}, \alpha_{\text{ideal}})$, the corresponding OGLCT is computed as $\hat{\mathbf{f}}_{\mathbf{M}_{\text{ideal}}} = \mathcal{F}_{\mathbf{M}_{\text{ideal}}} \mathbf{f}$. To minimize the discrepancy between the ideal and estimated OGLCT, the loss function is formulated as

$$\mathcal{R} = \frac{1}{MN} \|\hat{\mathbf{f}}_{\mathbf{M}_{\text{ideal}}} - \hat{\mathbf{f}}_{\mathbf{M}_{\text{est}}}\|_F^2 + \zeta(\xi^2 + \sigma^2 + \alpha^2), \quad (41)$$

where $\hat{\mathbf{f}}_{\mathbf{M}_{\text{est}}} = \mathcal{F}_{\mathbf{M}_{\text{est}}} \mathbf{f}$ denotes the estimated value, computed using the current parameters $\mathbf{M}_{\text{est}} = (\xi, \sigma, \alpha)$, and ζ represents the regularization parameter. To optimize the parameters ξ , σ , and α , the gradient of the loss function is:

$$\frac{\partial \mathcal{R}}{\partial \xi_k} = -\frac{2}{MN} \text{Re} \left(\text{Tr} \left((\hat{\mathbf{f}}_{\mathbf{M}_{\text{ideal}}} - \hat{\mathbf{f}}_{\mathbf{M}_{\text{est}}})^H (-i\pi Q^2 \mathcal{F}_{\mathbf{M}_{\text{est}}} \mathbf{f}) \right) \right) + 2\zeta \xi_k,$$

$$\begin{aligned}\frac{\partial \mathcal{R}}{\partial \sigma_k} &= -\frac{2}{MN} \operatorname{Re} \left(\operatorname{Tr} \left((\hat{\mathbf{f}}_{\mathbf{M}_{\text{ideal}}} - \hat{\mathbf{f}}_{\mathbf{M}_{\text{est}}})^H (-i\pi \frac{1}{\sigma_k} (\mathcal{P}\mathcal{Q} + \mathcal{Q}\mathcal{P}) \mathcal{F}_{\mathbf{M}_{\text{est}}} \mathbf{f}) \right) \right) + 2\zeta \sigma_k, \\ \frac{\partial \mathcal{R}}{\partial \alpha_k} &= -\frac{2}{MN} \operatorname{Re} \left(\operatorname{Tr} \left((\hat{\mathbf{f}}_{\mathbf{M}_{\text{ideal}}} - \hat{\mathbf{f}}_{\mathbf{M}_{\text{est}}})^H (-i\pi^2 \frac{\mathcal{P}^2 + \mathcal{Q}^2}{2} \mathcal{F}_{\mathbf{M}_{\text{est}}} \mathbf{f}) \right) \right) + 2\zeta \alpha_k,\end{aligned}$$

the quasi-Newton method [37,38] is employed. This iterative optimization approach avoids the direct computation of the Hessian matrix by constructing an approximation based on gradient information. The parameter updates within the quasi-Newton framework are described in Algorithm 2.

Algorithm 2 Quasi-Newton optimization workflow.

Input: Initial parameters $\xi_0, \sigma_0, \alpha_0$, convergence threshold ϵ , maximum iterations $K_{\max}$.

Output: Optimized parameters $\xi_{\text{est}}, \sigma_{\text{est}}, \alpha_{\text{est}}$.

- 1: Initialize $\mathbf{H}_{o,0} = \mathbf{I}$, $o \in \{\xi, \sigma, \alpha\}$ (initial Hessian inverses).
 - 2: **for** $k = 0, 1, 2, \dots$ **do**
 - 3: Compute gradients $\frac{\partial \mathcal{R}}{\partial \xi_k}, \frac{\partial \mathcal{R}}{\partial \sigma_k}, \frac{\partial \mathcal{R}}{\partial \alpha_k}$.
 - 4: Determine search directions $d_{o,k} = -\mathbf{H}_{o,k} \frac{\partial \mathcal{R}}{\partial o_k}$, $o_k \in \{\xi_k, \sigma_k, \alpha_k\}$.
 - 5: Perform line searches to find step sizes $\eta_{p,k}$.
 - 6: Update parameters $o_{k+1} = o_k + \eta_{o,k} d_{o,k}$.
 - 7: Compute updates: $s_{o,k} = o_{k+1} - o_k$, $y_{o,k} = \frac{\partial \mathcal{R}}{\partial o_{k+1}} - \frac{\partial \mathcal{R}}{\partial o_k}$.
 - 8: Update Hessian inverses [39,40]: $\mathbf{H}_{o,k+1} = \left(1 - \rho_{o,k} s_{o,k} y_{o,k}^T\right) \mathbf{H}_{o,k} \left(1 - \rho_{o,k} y_{o,k} s_{o,k}^T\right) + \rho_{o,k} s_{o,k} s_{o,k}^T$, where $\rho_{o,k} = \frac{1}{y_{o,k}^T s_{o,k}}$.
 - 9: **if** $\sqrt{\left(\frac{\partial \mathcal{R}}{\partial \xi_k}\right)^2 + \left(\frac{\partial \mathcal{R}}{\partial \sigma_k}\right)^2 + \left(\frac{\partial \mathcal{R}}{\partial \alpha_k}\right)^2} < \epsilon$ **or** $k \geq K_{\max}$ **then**
 - 10: **Break:** Convergence achieved.
 - 11: **end if**
 - 12: **end for**
 - 13: **Output:** Optimized parameters $\xi_{\text{est}}, \sigma_{\text{est}}, \alpha_{\text{est}}$.
-

Based on the graphs shown in Fig. 5, we analyze parameter optimization for two graph signals: $\mathbf{f}_3 \sim \text{Uni}(0, 1)$ on an undirected graph and $\mathbf{f}_4 \sim \text{Uni}(0, 1)$ on a directed graph. The ideal parameters for the undirected graph are $\xi_{\text{ideal}} = 0.21$, $\sigma_{\text{ideal}} = 0.66$, and $\alpha_{\text{ideal}} = 0.7$, while for the directed graph, they are $\xi_{\text{ideal}} = 0.51$, $\sigma_{\text{ideal}} = 1$, and $\alpha_{\text{ideal}} = 1.6$. Using the quasi-Newton method and a loss function, the optimization process iteratively refines the parameters for both cases. For the undirected graph, the parameters converge after 35 iterations to $\xi_{\text{est}} = 0.221$, $\sigma_{\text{est}} = 0.661$, and $\alpha_{\text{est}} = 0.696$, with a final loss of $\mathcal{R} = 1.302 \times 10^{-2}$. For the directed graph, convergence occurs after 19 iterations, with the parameters refined to $\xi_{\text{est}} = 0.506$, $\sigma_{\text{est}} = 1.006$, and $\alpha_{\text{est}} = 1.590$, and a final loss of $\mathcal{R} = 3.895 \times 10^{-2}$. These results demonstrate effective convergence in both cases, with the optimized parameters closely aligning with their respective ideal values. The outcomes highlight the robustness of the optimization method across different graph structures. Although the quasi-Newton method was adopted in our experiments for its balance between computational efficiency and convergence rate, the continuous parameter space of OGLCT is inherently compatible with alternative optimization strategies, such as stochastic gradient descent, Bayesian optimization, and genetic algorithms (Tables 1 and 2).

Table 1
Parameter optimization on undirected graph.

Iteration	ξ	σ	α	loss
1	0.000	1.000	1.000	1.158e+01
10	-0.012	0.752	0.685	1.419e-02
20	0.171	0.636	0.738	1.308e-02
30	0.226	0.663	0.693	1.302e-02
35	0.221	0.661	0.696	1.302e-02

Table 2
Parameter optimization on directed graph.

Iteration	ξ	σ	α	loss
1	0.000	1.000	1.000	8.442e-01
5	0.193	1.285	1.526	4.153e-02
10	0.423	0.983	1.631	3.905e-02
15	0.501	0.980	1.610	3.897e-02
19	0.506	1.006	1.590	3.895e-02

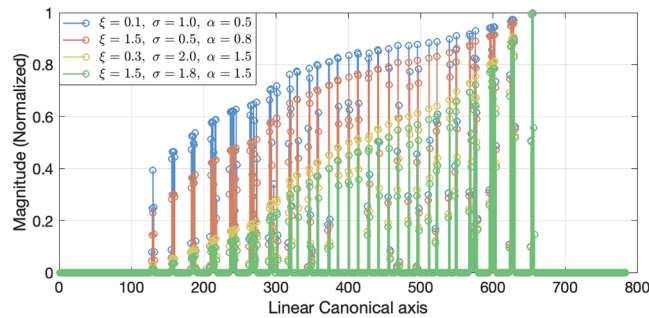


Fig. 8. Performance of OGLCT on the same image under different parameters.

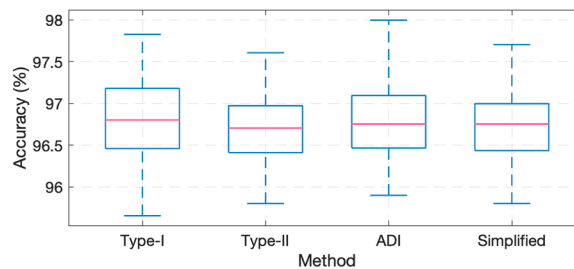


Fig. 9. Accuracy under different methods.

5. Application in image classification

In this section, the feature extraction capability of the OGLCT is investigated. By transforming graph signals from the vertex domain to the graph linear canonical domain, the OGLCT offers a flexible tool for analyzing graph-structured data. Using the MNIST dataset as an example, the image data are modeled as a graph structure, and the effects of different OGLCT parameters on the transformation results are explored. The MNIST dataset consists of grayscale images of handwritten digits, each with a size of 28×28 pixels. To apply the OGLCT, the images are modeled as graph signals, with each pixel corresponding to a vertex in the graph. Edges are defined between pixels based on their spatial adjacency (8-neighbor connectivity). The normalized OGLCT results for the four parameter sets are illustrated in Fig. 8. The results highlight the versatility of the OGLCT in extracting features under different parameter configurations. By adjusting ξ , σ , and α , the transformation can be tailored to focus on specific graph signal characteristics, offering a powerful tool for graph-based data analysis.

The OGLCT is then applied to a classic image classification task. The process involves utilizing the OGLCT for feature extraction, transforming graph signals from the vertex domain to the graph linear canonical domain. The extracted features are incorporated into a single-layer neural network with 8 hidden nodes for classification. Performance is evaluated under various parameter configurations (e.g., $\xi \in [0, 2]$, $\sigma \in [0.5, 2.5]$, $\alpha \in [0.5, 1.5]$) across four methods, highlighting the adaptability and efficacy of the approach. The MNIST dataset is utilized, and 6824 samples corresponding to the digits 4 and 9 are selected for the binary classification task. The classification results, illustrated in Fig. 9, reveal key observations regarding the four methods. Both Type-I and Type-II achieve a median accuracy of approximately 96.75%, with Type-II demonstrating slightly better stability owing to a narrower interquartile range (IQR). ADI outperforms the other methods, achieving the highest median accuracy of approximately 96.80% while maintaining a comparable IQR. The Simplified method achieves a slightly lower median accuracy of 96.70% but demonstrates the most consistent performance across parameter configurations, attributed to its limited variability.

For the selected method (Simplified), the results under different parameter configurations are illustrated in Figs. 10–12. The method demonstrates varying sensitivity to each parameter: ξ has the most significant impact, as extreme values (e.g., $\xi < 0.5$ or $\xi > 2.0$) result in reduced accuracy, whereas α exhibits the least sensitivity, maintaining stable performance across its tested range. Notably, the highest accuracy peaks (98.44%) are attained when $\xi = 1.9$, $\sigma = 1.6$, and $\alpha = 1.2$. The interplay between parameters underscores the importance of balanced tuning, as deviations from these ranges can significantly degrade performance. Additionally, traditional GFT, GFrFT, and GLCT are employed to extract features from images, and the results are presented in Fig. 13. GFrFT and GLCT also undergo parameter selection steps, where the parameters corresponding to the best-performing model results are chosen. For GFrFT, the parameter $\alpha = 0.5$ is used, while for GLCT, the parameters $\xi = 2$, $\sigma = 2$, and $\alpha = 1$ are selected. These results indicate that OGLCT can achieve exceptional accuracy under carefully tuned parameter configurations, highlighting its potential for high-performance applications.

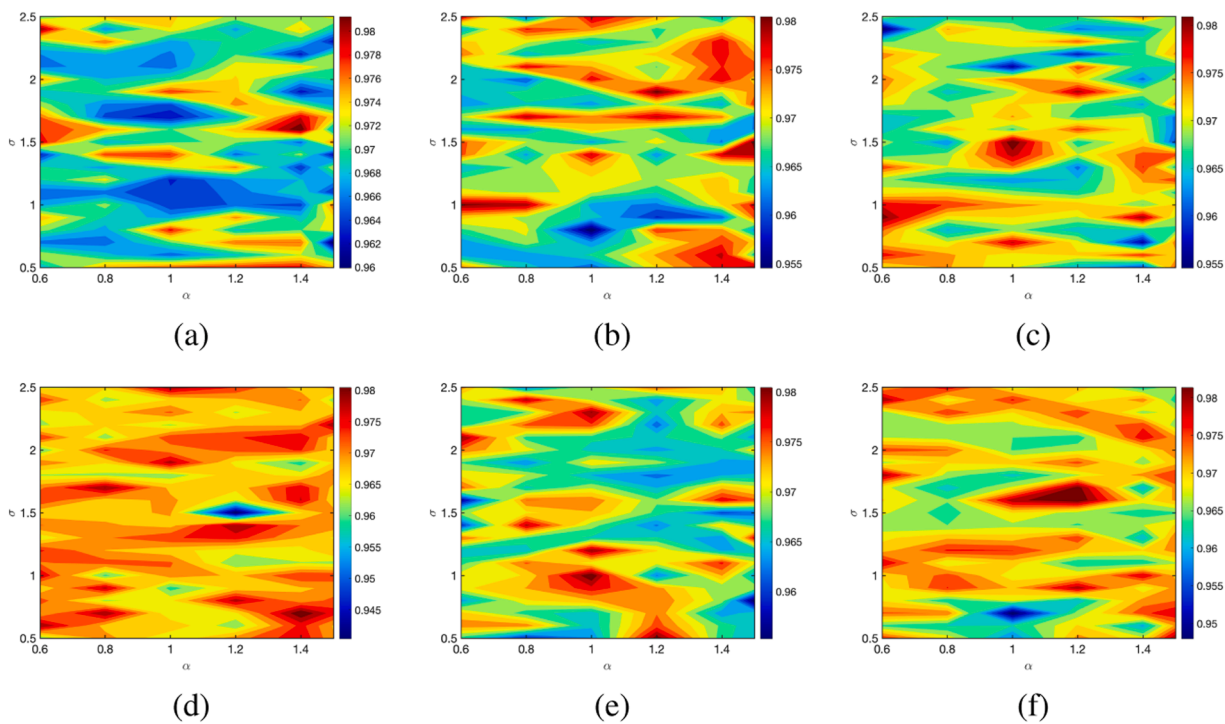


Fig. 10. Accuracy with different parameters. (a) $\xi = 0.1$. (b) $\xi = 0.4$. (c) $\xi = 1$. (d) $\xi = 1.1$. (e) $\xi = 1.6$. (f) $\xi = 1.9$.

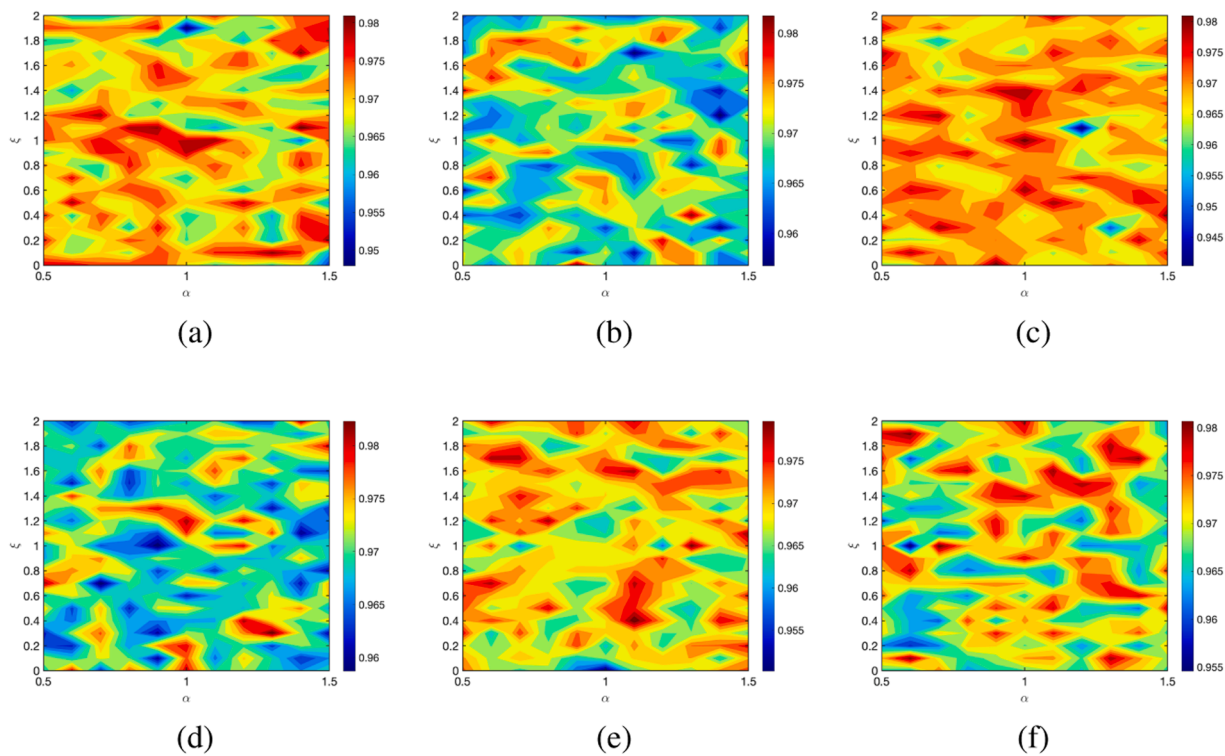


Fig. 11. Accuracy with different parameters. (a) $\sigma = 0.7$. (b) $\sigma = 1.3$. (c) $\sigma = 1.5$. (d) $\sigma = 1.9$. (e) $\sigma = 2.2$. (f) $\sigma = 2.4$.

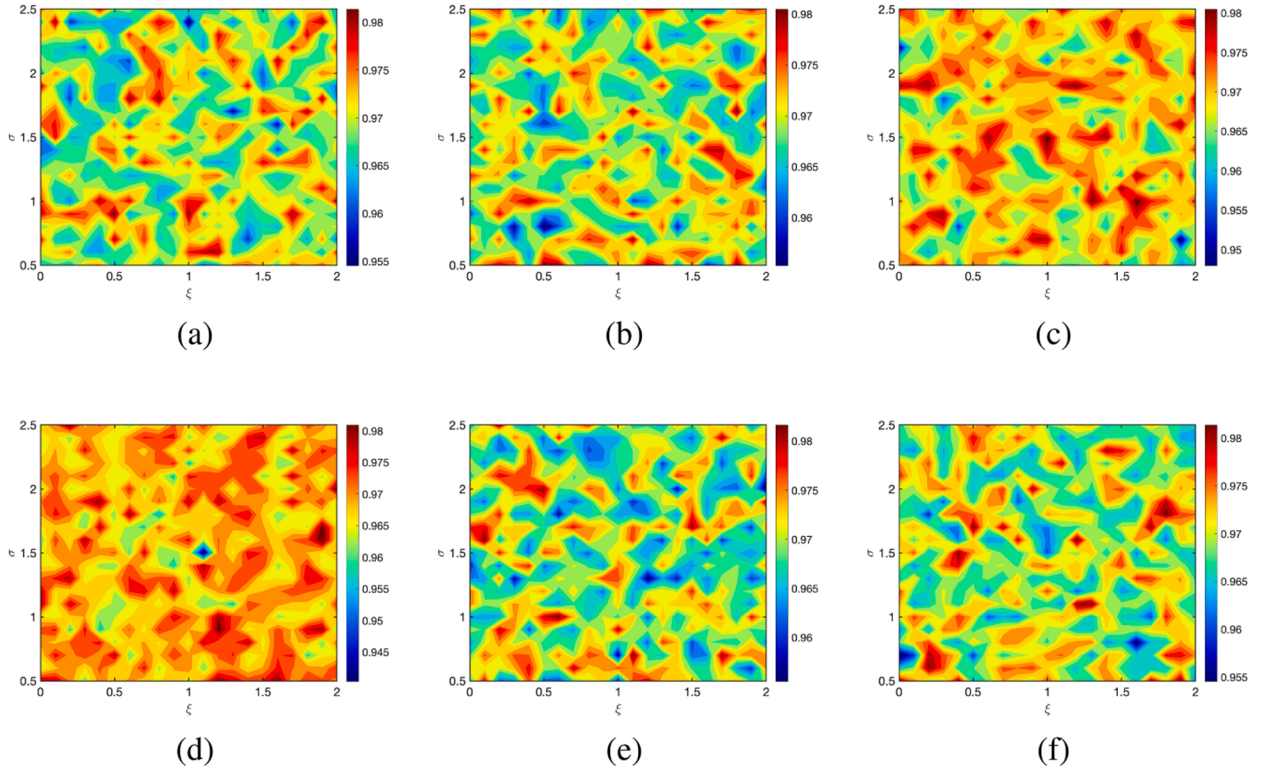


Fig. 12. Accuracy with different parameters. (a) $\alpha = 0.6$. (b) $\alpha = 0.8$. (c) $\alpha = 1$. (d) $\alpha = 1.2$. (e) $\alpha = 1.4$. (f) $\alpha = 1.5$.

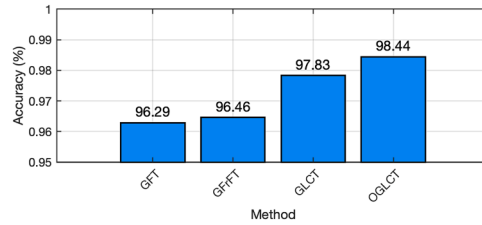


Fig. 13. Accuracy with different spectral graph transforms.

6. Conclusion

An operator-based graph linear canonical transform (OGLCT) framework has been proposed, unifying three fundamental graph signal transformations via hyper-differential operators. By solving the Sylvester equation to derive the core operators $\mathcal{P}$ and $\mathcal{Q}$, a rigorous mathematical foundation is established, intrinsically integrating graph fractional Fourier transforms, scaling operations, and chirp modulation. This operator-based approach eliminates the stage-wise spectral recomputations required in previous methods, thereby ensuring a unified mathematical formulation across all transformations through shared operator dynamics.

To further elucidate the practical utility of the framework, a systematic analysis has identified three complementary computational methodologies: (1) explicit solutions provide theoretical foundations through closed-form formulations; (2) ADI iterations facilitate scalable processing for large graphs via efficient matrix splitting; and (3) spectral decomposition supports stable implementations through eigenvalue-based simplifications. This methodological spectrum achieves a synergistic balance-offering theoretical verification via analytical tools, practical scalability through iterative adaptation, and computational efficiency grounded in spectral structure. In addition, the framework's parametric adaptability enhances its flexibility across a wide range of graph structures.

The proposed approach has been empirically validated through image classification case studies, demonstrating significant performance improvements achieved via OGLCT-based feature embedding. These benefits are particularly evident when processing non-Euclidean structures such as social networks and sensor arrays. Notably, the ability to customize time-frequency resolution and enhance spectral localization offers substantial advantages over conventional graph Fourier methods, effectively addressing limitations associated with non-stationary signals and varying vertex-frequency relationships in irregular domains.

Looking ahead, future research is expected to advance along three key directions: (1) the design of graph neural network architectures optimized for OGLCT; (2) the construction of vertex-frequency distributions for joint time-vertex analysis; and (3) the extension of the framework to dynamic hypergraph structures. The mathematical foundation established in this work offers a robust and versatile platform for advancing graph signal processing in increasingly complex and irregular domains.

CRedit authorship contribution statement

Jian-Yi Chen: Visualization, Methodology, Writing – original draft, Validation; **Bing-Zhao Li:** Supervision, Writing – review & editing; **Linyu Peng:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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